Policy priorities for stablecoin regulation: past, present and future

Financial Applications and Social Economics Working Group

BGIN

Working Draft

Version: 04 June 2025

Instructions for contributors

bg in

blockchain governance initiative network

The Global Network for Blockchain Stakeholders™

©2024 BGIN (Blockchain Governance Initiative Network) is registered as BN Association, a Japanese general association with its principal place of business at Shiba Building, 704, 4-7-6, Shiba, Minato-ku, Tokyo. All rights reserved.

Policy priorities for stablecoin regulation - past, present and future
A comparative analysis of MiCA (European Union) and VARA (United Arab Emirates)

Chloe White
Co-Chair, Financial Applications & Social Economics Working Group
Blockchain Governance Initiative Network (BGIN), Tokyo
26.03.25

Abstract

Countries have increasingly recognized the importance of stablecoin regulation in recent years, leading to significant policy and legislative action in diverse jurisdictions. This paper offers a comparative analysis of two new regulatory regimes for stablecoins: those of the European Union and the United Arab Emirates. These two regions both present particularly compelling case studies due to the wider context of their comprehensive virtual asset regulations, which cover a broad range of activities and services; as well as their newness, being implemented recently or imminently. In examining these two fulsome and modern stablecoin regimes, we share insights into the diverse drivers and policy motives behind each set of virtual asset regulations. We observe that these policy drivers are relevant across borders and are likely to arise in future regimes in other jurisdictions. The paper begins with an overview of the key events that catalysed regulators' decision-making on their policy priorities, which provides critical context to the current discourse on stablecoins. Furthermore, the analysis identifies potential policy gaps and issues that may require further examination by practitioners and policymakers. We conclude by providing a forward-looking perspective on anticipated legal reforms in the realm of stablecoins, including proposed reforms that are now emerging in the United States.

1. Pivotal moments in stablecoin policy: key events that shaped present-day regulation

1.1 Blockchain payments in their first decade: 2008-2018

A starting point for considering developments in this sector may be to ask, why have regulators bothered to contend with stablecoins at all, given other pressing policy priorities? After the invention of Bitcoin in 2008 and its subsequent launch in 2009, interest in virtual assets has grown by orders of magnitude – whether measured by the number of projects, wallets, investors, developers, or financial or other metrics. However, while many critics of this era questioned the sustainability of the virtual asset industry as anything more than a passing fad, the highly volatile sector continued to grow despite its challenges and, accordingly, to reveal its inherent risks. Indeed, many of these risks have materialised in a series of high-profile industry failures: notably the implosion of the Initial Coin Offering (ICO) movement, which peaked in 2017 and which was primarily focused on crowdsourced retail funding models; and the great crypto credit crunch of 2022, which featured around the

dramatic collapse of a popular stablecoin project on 9 May 2022, and culminated with FTX's declaration of bankruptcy on 11 November 2022, discussed later in this section.

Prior to the ICO boom, most of the regulatory work that had occurred in the virtual asset sector had been largely limited to: (1) ensuring that cryptocurrency exchanges could be captured by anti-money laundering rules, and (2) to ensuring that the capital gains realised by fortunate investors would be captured under tax law and enforcement activities. In some cases, the legal community also moved to clarify the application of property laws to Bitcoin. Some government programs also encouraged experiments with blockchain technology through innovation labs, or pilot projects. However, many of the early mover jurisdictions that aimed to create bespoke frameworks, designed to accommodate token issuance, were smaller nations – such as Estonia, Lithuania, Malta and Gibraltar, along with Switzerland, which became home to 'Crypto Valley' boasting 14 crypto unicorns (source). It was not until 2018 that virtual asset policy was elevated to G20 communiqué status.

The industry's best-recognised stablecoins all launched at different times during this era, and are all pegged to the US dollar. Tether (USDT) launched in 2014 under the name 'Realcoin' and remains the dominant stablecoin today by market capitalisation. DAI, a decentralised stablecoin managed by MakerDAO, launched in late 2017 with Ether (ETH) as its backing collateral. The DAO later converted DAI to a multi-collateral system and introduced a savings rate (DSR). DAI has the third-larget market capitalisation, behind USDC, which was first issued by Circle in 2018, and which claims to be the dominant regulated stablecoin globally. Despite USDT's consistent ranking as the largest stablecoin by market capitalisation, Tether has faced strong scrutiny from broad stakeholder groups, and remains a regular source of controversy among industry commentators.

With respect to financial services laws, we observed that, during this era, regulators in larger economies initially tended towards the release of guidance notes and were slower to work on direct legislative reforms (for example, Australia and the United States). Throughout this period, most policymakers remained unconvinced that Bitcoin and other cryptocurrencies could play a meaningful role in global payments, given the dominant role of fiat, as well as scepticism about blockchain scalability. However, while the rise of Bitcoin and the ICO boom may not have been sufficiently motivating for larger governments to want to prioritise financial services law reform, the emergence of stablecoins eventually proved more convincing, sparking another substantive wave of policy attention globally, including in larger economies across Europe and the UK, and the Asia-Pacific.

1.2 The rise of stablecoins in a tumultuous era: 2019-2022

Stablecoin policy discussions first gained serious momentum in the regulatory community upon Facebook's announcement of its Libra stablecoin project in June 2019. This initiative, while ambitious, sparked immediate and intense backlash from various governments and regulatory bodies worldwide. Concerns over financial stability, monetary sovereignty, and consumer protection (including data privacy) led to significant opposition globally; most vocally in European Union nations. This international backlash ultimately dissuaded the consortium, headquartered in Switzerland, from proceeding with its initial plans. As a result, the project faced numerous hurdles and eventually rebranded to Diem before being disbanded entirely in early 2022.

コメントの追加 [1]: It was only in 2018 that its market cap boomed, though.

コメントの追加 [2]: FYI the Genesis Act, introduced in Feb 2025 in the US, requires the Treasury, in consultation with federal agencies, study the risks of stablecoins that rely on self-referential collateral (e.g., algorithmic stablecoins like Terra/Luna)

コメントの追加 [3]: https://www.banking.senate.gov/newsroom/majority/scott-hagerty-lummis-gillibrand-introduce-legislation-to-establish-a-stablecoin-regulatory-framework

Regulatory debates continued throughout this period, and many policy professionals (including the Bank for International Settlements) asserted that Central Bank Digital Currencies (CBDCs) would become important antidotes to the cryptocurrency craze, as well as offering interesting new tools to support governmental goals related to payments and financial system management. The sitting Bank of England Governor, Mark Carney, discussed the potential benefits of fiat currency tokenisation in his final speech at the Jackson Hole Economic Symposium in August 2019. He proposed that a Synthetic Hegemonic Currency (SHC) provided by a network of central bank digital currencies could reduce reliance on the U.S. dollar and help mitigate the negative impacts of global financial cycles and trade tensions (source). Within the G20, stablecoins first began to feature as part of cryptocurrency conversations in 2019, were then first mentioned explicitly in its communiqué in 2020, and has remained a prominent topic in all subsequent releases.

The 'everything bubble' that resulted from COVID-related shocks and fiscal stimulus left few markets untouched, and virtual assets were no exception.[1] The on-chain yield-farming frenzy known as 'DeFi Summer' kicked off in earnest in 2020, and stablecoins performed an important role in the DeFi ecosystem, including as collateral for on-chain loans and liquidity provision, on platforms such as Uniswap. Following the peak of DeFi Summer in 2020, attention turned to NFTs, which experienced their peak of frenzy in 2021.[2] Bitcoin then reached its cycle peak in November 2021. While these were thrilling events for participants on the frontlines, these occurrences did not meaningfully impact the direction or urgency of virtual asset policy work that was underway globally. Primarily because complex 'Layer 2' activities would not emerge onto the radar of decision-makers until a few years later, and because COVID remained the policy priority at this time.

However, the virtual asset sector subsequently experienced a series of significant governance failures throughout 2022, catalysing urgent and widespread discussions about the need for regulatory oversight across the industry, and pressed again on concerns about stablecoins and their inherent risks. In particular, the wealth destruction caused by Terra-Luna's downward spiral raised alarms about the stability of algorithmic stablecoins. The resulting credit and liquidity crunch permanently re-shaped the industry and raised concerns about systemic risk in the sector. In particular, the collapse of Celsius, and the revelations about FTX's improper management, heightened concerns about the adequacy and appropriateness of backing consumer liabilities with self-issued virtual asset tokens. These events also made clear the urgent need for better corporate governance and oversight in the sector, giving momentum to some of the industry's self-regulatory ideas such as 'Proof of Reserves', and focusing attention on the importance of asset segregation, sound treasury management practices, and even calling into question the role and responsibility apportioned to the venture capital community that had aggressively backed these projects.[3]

---

[1] See https://en.wikipedia.org/wiki/Everything_bubble.

[2] Significant transactions that occurred at this time included the sale of a digital artwork by Beeple (Mike Winkelmann) at Christie's auction house for $69 million in March 2021, and Visa's purchase of CryptoPunk #7610 for 49.5 ETH in August 2021.

[3] Exposure to Terra-Luna proved to be catastrophic for a number of high-profile businesses. Most notably, Three Arrows Capital blew up as a result of its UST exposure and filed for bankruptcy in July 2022. This directly led to the collapse of Voyager Digital, which filed for bankruptcy in the same month, and also created liquidity issues for BlockFi, which filed for bankruptcy later in the year.

Of relevance to this paper's comparative analysis, it was during this era that the Government of Dubai passed cabinet legislation to authorise the creation of the Virtual Assets Regulatory Authority (VARA), the world's first crypto-dedicated regulatory agency. The "Dubai Virtual Asset Regulation Law" (Law No. 4 of 2022) was enacted in March 2022, and a comprehensive set of regulations and rulebooks were released around twelve months later. During this era the European Union also began consultations on its MiCA legislative package, or "Markets in Crypto-Assets (MiCA)", which was officially proposed by the European Commission in September 2020, and which entered into force in June 2023.[4]

Finally we note that critiques of Tether ramped up in this era. Ongoing complaints have primarily revolved around the accuracy of its disclosures, particularly with respect to the composition of its reserves; its third-party audit practices; and the relationship between Tether and its associated exchange Bitfinex, which was the subject of a legal case brought forward by the New York Attorney General in 2019 and, upon settlement in 2021, resulted in a fine of $18.5 million (source). Tether has made a number of public statements emphasising its intention to improve its practices but admitting no wrongdoing.

1.3 The present day, an era of renewed promises: 2023 onwards

The collapse of Silicon Valley Bank in 2023 created ripples for USDC and many other stablecoins, both centralised and decentralised. This event validated existing worries about consumer protection and systemic risk in this asset class.[5] However, by the time this incident occurred, policy responses to the events of 2019 and 2022 were already well-progressed (or implemented) in regions such as the UK, EU, Middle East and Asia. Risks surrounding the quality of backing collateral, the need for strong governance, and the importance of transparency and audits, were already understood in 2023 following the more catastrophic incidents of the prior year. Furthermore, while the USDC de-peg incident revealed important and interesting revelations about the potential systemic linkages between traditional finance and virtual assets, it's issuer Circle - a company licensed within the United States - was not accused of any wrongdoing, and USDC recovered well from the incident, returning to its 1:1 dollar peg, which it has subsequently maintained.

Due to all of these factors, in relative terms, the USDC de-peg event did not have as strong of an impact on stablecoin policy debates as did the events of 2019 and 2022. Although USDC's market capitalisation remains smaller today than it was prior to the de-peg incident, USDC is now available via more than a dozen Layer 1 and Layer 2 protocols, and the stablecoin sector as a whole has enjoyed substantial growth. Tether's USDT remains the leading product in the stablecoin space, having achieved a historic market capitalisation of $111 billion in May 2024 (source) after first breaching the $100 billion mark two months prior, as reported by Cointelegraph based on data printed by CoinGecko (source).

コメントの追加 [4]: blockchains

[4] Provisions related to asset-referenced tokens (ARTs) and electronic money tokens (EMTs) apply from 30 June 2024.
[5] A thorough evaluation of the spectrum of risks revealed by this depegging incident has been published by BGIN (see Kakebayashi 2023).

Despite the many challenges that the industry has faced, the stablecoin business model has sustained particular relevance over time and is now enjoying an era of renewed promise. In April 2024, Visa published on their website an analysis revealing “steady growth” in regular users of stablecoins “with 27.5 million monthly active users across all chains” (source). This has taken place against a backdrop of wider successes for the industry, earned over the past twelve months, including: the launch of several spot ETF products for Bitcoin in the United States, the validation of tokenised financial products by BlackRock, and PayPal launching its own stablecoin.

(reference to Potts Berg White 2024).

Given that few products in the space have proven to be more successful at this point in the industry’s development, it is clear to see why regulators have begun to develop stablecoin-specific rules for the industry to follow, particularly when taking into account the clear risks posed to consumers and to financial stability. These risks have been formally identified within multilateral policy organisations, and dealt with first-hand by market participants.

2. The impact of recent events on regulator priorities

In response to the above events, regulators worldwide began advancing numerous policy initiatives to address these concerns, some of which remain in debate and/or consultation phase, and some which have progressed to implementation. The most pressing priorities for consumer protection and market integrity identified from these incidents included several key areas of regulation:

1. Rehypothecation of assets,
2. Audits and other disclosures,
    a. including rules for White Papers,
3. Segregation of customer assets,
    a. and proper governance of the reserve, including liquidity management, and
4. Prudential and risk management.

Additionally, broader governance issues, such as preventing and managing conflicts of interest, also came to the forefront.

コメントの追加 [5]: This sentence seems to be missing a word or have an extra word.

These issues can be seen as clearly reflected in two jurisdictions that have recently brought new stablecoin rules into force: the Emirate of Dubai, and the European Union. VARA’s stablecoin issuance policy came into effect upon its publication in 2023, while the EU’s specific provisions regarding asset-referenced tokens and e-money tokens enters into force on 30 June 2024. What follows is an exploration as to how the above policy priorities have been interpreted and mitigated under each regime.

3. Comparing stablecoin rules under MiCA in the EU vs. VARA in Dubai

Overall, the stablecoin regulatory regimes in the EU and Dubai are very similar, focusing on the same policy thematics, and with comparable mitigations and restrictions. However, there are some differences in how the policies are operationalised.

3.1 Rehypothecation of assets is prohibited by default

Rehypothecation, which refers to the practice of reusing collateral posted by clients, became a major concern in 2022. Rehypothecation practices are tightly regulated under both MiCA and VARA, reflecting a shared concern over the potential misuse of reserve assets.

Article 45(4) of MiCA explicitly states that reserve assets must not be rehypothecated or encumbered, and any such practices must be fully disclosed in the whitepaper. Similarly, VARA restricts rehypothecation of reserves backing stablecoins, ensuring these assets are not used for other purposes. Any exceptions to this rule must be disclosed in the whitepaper and approved by the regulator.

3.2 Audits are compulsory and require third party assurance

Under both MiCA in the European Union and VARA in Dubai, stablecoin issuers are subject to stringent audit requirements, though the specifics vary. The primary risk these regulations aim to mitigate is the potential insolvency of stablecoin issuers, ensuring they have sufficient reserves to meet redemption requests and maintain market confidence. Both regimes require issuers to disclose audit results and reserve holdings publicly, enhancing transparency and trust within the market.

MiCA mandates that issuers of asset-referenced tokens conduct periodic audits, typically on an annual basis, to verify the adequacy and quality of the reserves backing the tokens. These audits are detailed in Article 45(2), ensuring that statutory auditors or audit firms oversee the process.

Similarly, VARA requires issuers of Fiat-Referenced Virtual Assets (FRVAs) to undergo regular audits, which include internal and independent third-party audits. These audits confirm the existence, value, and accessibility of the reserves, ensuring compliance with regulatory frameworks.

3.2a White Paper requirements specify the details that must be disclosed to the public

Early in its existence, the cryptocurrency community converged on a common practice of publishing various disclosures via 'White Paper' publication, which is sometimes compared to a Product Disclosure Statement usually found in traditional financial services. The practice of issuing a White Paper has been endorsed in Europe and in Dubai, including in the case of stablecoins.

コメントの追加 [6]: On MiCA, things that might be worth calling out as interesting -

1. the additional requirements relating to non-Euro stablecoins - monitoring of usage as means of payment and associated caps.

2. significant stablecoins - additional reserve assets - impact on profitability

More broadly I think there's value expanding the aperture to look at other jurisdictions.

- Treatment of unregulated stablecoins - whether they are permitted for retail listing and trading

- Permissioned vs. permissionless chains - e.g. in Japan I understand there is significant cautiousness about permissionless chains

コメントの追加 [7]: I am also curious if there is any / potential difference in term of eligible asset - for example government MMF is allowed, any maturity, concentration requirement?

コメントの追加 [8]: I am curious about reverse repo practices and to what extent this is allowed in various countries. Only CCP cleared repo and how long is the maturity?

コメントの追加 [9]: Good point. It would be worth discuss in WG meetings as well as in Block #12 whether to include comparative analysis of regulations on
(1) asset (fiat) backed stablecoins; and
(2) algorithmic stablecoins
on a sigle paper or in a series of papers.

As we may aware MiCA defines
(1) e-money tokens and asset-referenced tokens (not using word "stablecoins"); and
(2) tokens other than EMT and ART (more stringent requirements)

Precedingly, Japan revised Payment Services Act in 2023, enacted in 2024, defining
(1) electronic payment instrument, and
(2) kinds of Virtual Asset (I assume it would be more stringent as well)

コメントの追加 [10R9]: Otherwise , we can discuss extended research on extraterritorial use of stablecoins, both financial stability and money laundering risk.

コメントの追加 [11]: This article refers to significant asset referenced tokens. Art 37 (1) a describes the same requirement for non-significant asset reference tokens.

コメントの追加 [12]: I don't find the same requirement on e-money tokens in MiCA. I assume similar requirements are stated in e-money regulations.

The content and approval process for whitepapers under MiCA and VARA exhibit several similarities aimed at ensuring comprehensive disclosure and regulatory compliance. MiCA stipulates that whitepapers must include detailed information about the token, its underlying assets, the mechanism for maintaining its value, governance structure, and risk disclosures, as specified in Articles 17 and 18. The whitepaper must be submitted to and approved by the national competent authority before issuance.

VARA's requirements are similarly rigorous. The whitepaper must detail the nature of the stablecoin, its backing assets, mechanisms for maintaining its peg, and rights and obligations of token holders. VARA also mandates the disclosure of material information impacting the token's value and the issuer's ability to maintain the peg.

These requirements form part of a range of policies within each regime that collectively aim to mitigate the risks that arise from a lack of transparency, and ideally will work to reduce information asymmetry in the market for stablecoins.

3.3a Client asset segregation is enforced alongside reserve asset management policies

Both MiCA and VARA provide detailed regulations on the management of reserve assets, focusing on segregation and accessibility. Among other goals, these measures are intended to mitigate liquidity risks and protect the reserve assets from being compromised, ensuring they are always available to back the stablecoins.

MiCA requires that the reserve assets be legally segregated from the issuers' estate and from the reserves of other asset-referenced tokens. Article 45(2) ensures that creditors of the issuers have no recourse to the reserve assets, particularly in the event of insolvency. Operational segregation is also mandated to ensure that the reserve assets remain dedicated to backing the tokens.

VARA mandates that reserve assets be maintained with financial services firms approved during the licensing process and segregated from the issuer's own funds. The assets must be accessible and convertible into reference currencies to process redemption requests promptly (more about liquidity is covered in the following section). Regular risk assessments are required to evaluate the appropriateness of the reserve assets' composition, ensuring diversification and regulatory compliance.

3.3b Liquidity and reserve policies emphasise responsiveness to redemption requests

Maintaining liquidity in backing assets was identified as crucial for ensuring stablecoins can meet redemption demands. Ensuring liquidity and efficient redemption processes is a critical focus for both MiCA and VARA. MiCA specifies liquidity requirements, including the types and maturities of assets that should be held to meet redemption requests. Articles 45 and 46

detail the requirements for maintaining sufficient liquidity to address redemption demands promptly. Further detail on liquidity requirements is forthcoming from the EBA.

VARA also emphasises liquidity, requiring issuers to maintain sufficient reserves to ensure that FRVAs are 100% backed at all times. The relevant rulebook mandates that redemption requests must be processed within one working day, barring significant disruptions. Policies must be in place to ensure timely access to reserve assets for redemption purposes.

3.4 Prudential requirements and risk management

Both regulatory frameworks emphasise the importance of high-quality collateral and stringent prudential standards to safeguard the stability of stablecoins. These requirements are designed to mitigate broader financial risks, ensuring that issuers have sufficient liquidity to meet redemption requests and maintain market stability, to prevent a recurrence of a depeg similar to the run on USDC that occurred in 2023.

MiCA allows a range of high-quality assets to be used as collateral, including cash and government bonds. Article 45 specifies that these assets must meet certain liquidity and credit quality standards, ensuring they can be liquidated rapidly with minimal adverse effects. Issuers must maintain adequate reserves and comply with specific liquidity requirements to meet redemption demands.

VARA also sets out specific standards for the types and quality of assets that can be used as collateral. Acceptable collateral includes cash, high-quality liquid assets (HQLA), and short-term government bonds. The assets must be highly liquid and low-risk, capable of covering the value of the issued tokens at all times. VARA's rulebook emphasises maintaining a percentage of reserves in highly liquid assets and implementing robust risk management frameworks.

3.5 Prohibition of Interest and Fees

MiCA and VARA both prohibit issuers from granting interest or benefits linked to the length of time holders possess the stablecoins. MiCA's Article 36 explicitly prohibits issuers and crypto-asset service providers from granting interest or any remuneration equivalent to interest. Similarly, VARA forbids issuers from providing any payments or benefits to incentivize the acquisition or holding of FRVAs.

Additionally, both regimes mandate that the redemption of stablecoins should not be subject to fees. This prohibition ensures that holders can redeem their tokens at par value without incurring additional costs.

These prohibitions are designed to mitigate risks associated with speculative investments and ensure that the stablecoins maintain their intended function as a stable medium of exchange rather than an investment vehicle.

コメントの追加 [13]: The Financial System Council of Japan started discussing making it more flexible. We discussed made a more flexible capital adequacy framework at Block#11, which I can share with you individually.

コメントの追加 [14]: the definitions of high-quality assets might be different across jurisdictions? It's worth noting here

コメントの追加 [15]: I didn't know about this. And from my understanding, it is not what is required under the Japanese law. It is interesting if the requirement is just for the purpose of maintaining the intended function as a stable medium of exchange. A lot of DeFi protocols give liquidity provision rewards to stablecoin holders anyways. It is already use also as an investment vehicle. Plus, while it is related to my paper on staking, retaking protocols started accepting stablecoins, which one of the examples that characterizes the trend of stablecoins being used as an investment vehicle.

コメントの追加 [16]: Such provisions are not logically consistent with broader practices in the financial world.

For example: why can a bank offer a savings account that pays interest on deposits of USD in this account whereas a crypto-asset service provider cannot offer, let's say, a staking vault that offers rewards for people who deposit USD-pecked USD-backed stablecoins in this vault?

Also, fiat currencies are not stable in the long-term precisely because of the "money printing / base interest rate" vicious cycle. To build a stablecoin product that is truly stable in the long-term, it is necessary to pass forward the base interest rate to stablecoin holders.

It is unclear what "remuneration equivalent to interest" means. For example, would a stablecoin that is pegged to USD from 2024 and thus indirectly protects its holders from inflation be prohibited? Would such a protection from inflation be considered a "remuneration equivalent to interest"?

コメントの追加 [17]: This is actually problematic for stability.

コメントの追加 [18R17]: ... and sustainability.

コメントの追加 [19]: I understand where these requirements are coming from - FSB high level recommendations - such as "without undue cost" "any fee should not be high". Does this mean that stablecoin issuers cannot adopt swing price or redemption fees?

4. Conclusion

In summary, both MiCA and VARA provide comprehensive regulatory frameworks to ensure the stability and integrity of stablecoins. While there are nuanced differences in their approaches, both regimes share common goals of enhancing transparency, maintaining adequate reserves, and protecting investors. The stringent audit, disclosure, and collateral requirements under both regulations aim to mitigate financial, operational, and liquidity risks, ensuring the robustness of the stablecoin market. [Both regimes also recognise the importance of AML/KYC compliance and procedures, although this is not limited to stablecoins.] [expand this section]

5. Future considerations

[short paragraph]

[writing about why we chose these considerations and what may come of these]

What issues still need to be addressed in the stablecoin space? [ideas] [chainalysis]

5.1 U.S. Policy Developments & Global Implications

The United States is now moving toward a comprehensive framework after several years of debate.

GENIUS Act would still prohibit yields. Note: Coinbase currently advertises 4.1% “rewards” on USDC on their website (18.6.25) rather than “yield” or “interest”

Note Circle IPO.

Castle Island Ventures stablecoin paper 2025

コメントの追加 [20]: I suggest putting "3.6 Conclusion" or "4. Conclusion", and describe in further detail what are the findings and implications to other jurisdictions by receiving the key expressions from "abstract".

We need to further discuss what are the key implications from the comparison and draw messages here even though some implications are already written in abve body.

We may find some of the key concepts of MiCA (and potentially VARA regulations) out of date and need revision as we observe the significant issuance share of USD denominated stablecoins in the market.

コメントの追加 [21]: I suggest using other expression in this title;
e.g. 1 "Perspectives for further studies".
e.g. 2 "Future considerations"
Reason: (1) We discussed in the WG calls that that we list issues yet contained in MiCA and VARA regulation, (2) These issues need to further examined prior to concluding policy considerations, (3) This section contained variety of "so-called stablecoins" as opposed to legally defined "tokens" in MiCA and VARA regulations in previous section as main theme of this paper.

コメントの追加 [22]: MiCA and VARA seem to be suitable for fiat-backed stablecoins... But there are many other types of stablecoins that are not properly covered.

Also, existing regulations seem to have a shallow and prejudiced understanding of what "algorithmic" means.

コメントの追加 [23]: A few issues that immediately come to mind:
1. Cross-border supervision - we have cross-border supervisory frameworks for banks, but not yet anything similar for international stablecoins.

2. Financial integrity - no clear consensus over stablecoin issuers' AML obligations - e.g. with regard to freeze, burn, blacklist capabilities, voluntary freezes of illicit assets etc.

3. Cybersecurity - has not been a big focus thus far of regulation. But stablecoins are token smart contracts. What happens if someone hacks and mints a whole bunch and that leads to a depeg with spillover implications?

More broadly, access to payment services, central bank accounts, etc.

コメントの追加 [24]: This may be included in "4.Conclusions"

APPENDICES

A.1 Yield-bearing Tokens

A notable recent development is the emergence of yield-bearing tokens, which share with users the interest earned on their reserve assets. Some of them brand themselves as “Yield-bearing Stablecoins”. Traditional fiat-backed stablecoins like USDT and USDC currently pay holders no yield, even though their issuers generate significant income from investing reserve funds. With U.S. interest rates elevated (around 4–5% in 2024–25), users collectively forgo over $9 billion in annual interest by holding non-yielding stablecoins. This foregone income has catalysed innovation: new tokens such as:

- USDM (offered by Mountain Protocol)
- Ondo’s USDY or OUSD
- AGDM

These novel tokens are designed to pass through treasury bill interest or DeFi lending yields to the coin holders.

These yield-bearing coins essentially function like tokenized money market funds, providing a stable USD peg plus a return, and have quickly gained traction. By mid-2025, yield-bearing stablecoins in circulation reached about $11 billion (roughly 4.5% of the total stablecoin market), up from only $1.5 billion a year earlier (https://cointelegraph.com/news/yield-bearing-stablecoins-hit-11b-pendle-dominates-growth)

We will need to carefully consider the economic implications of yield-bearing tokens. On one hand, these products can make coins more attractive to hold (improving “store of value” functionality) and potentially draw in more retail and institutional adoption. On the other hand, if not properly supervised, they could introduce new risks: liquidity mismatches if assets held to generate yield are less liquid, or even competitive pressure on banks if large amounts of money move from bank deposits (which are tightly regulated and insured) into such coins offering higher returns.

Note from GBBC meeting: “Bermuda has licensed quite a number of yield bearing stablecoin issuers under DABA”

A.2 Ongoing Concerns with Algorithmic Stablecoins

Even as collateralized stablecoins evolve, the industry has not fully given up on algorithmic stablecoins – those that attempt to maintain a peg purely through algorithms and market incentives rather than traditional assets in reserve. However, from a policy perspective, these designs remain highly problematic. There is inherent fragility to algorithmic stablecoin models, which can enter death spirals when confidence falls and the technology is unable to sustainably maintain economic balance between mint and burn functions.

Unfortunately, the sins of algorithmic stablecoins have stained the reputation of stablecoins, financial authorities now frequently cite the failure and collapse of algorithmic stablecoins as

コメントの追加 [25]: I suggest using "Yield-bearing Tokens"
Reason: It seems the Term Yield-bearing Stablecoin is used as a marketing word by Mountain Protocol (ond others if any). As explained in the body of this section, the aim of this type of tokens are to preserve the value of asset in the inflationary market. Users primarily need to convert into USDT, USDC, and their likes to transact. This is very much like the development of personal checks linked to MMFs in the US market. Management companies of MMFs have made alliance with (small) banks so that they can redeem the exact amount of fund to clear the personal check by receiving bank. Similar developments occurred for corporate accounts as well.

コメントの追加 [26R25]: Resolved 18.6.25

コメントの追加 [27]: Ondo carefully avoid using the word "Yield-bearing stablecoins", instead defining as "yield-bearing token secured by US Treasuries", even though giving an impression as stablecoin by stating " Enjoy the utility of stablecoins, now with safe daily yield."

(legal statement)
USDY is not, and may not be, offered, sold, or otherwise made available in the US or to US persons. USDY has not been registered under the US Securities Act of 1933, as amended ("Act") or pursuant to securities laws of any other jurisdiction, and may not be offered, sold or otherwise transferred in the US or to US persons unless registered under the Act or an exemption or exclusion from the registration requirements thereof is available.

コメントの追加 [28R27]: Resolved 18.6.25

コメントの追加 [29]: It could contain many implications. For example, if a coin (token) is licensed in Bermuda, does it mean it it is lincensed to market to individuals/institutions in other jurisdiction? This could resemble the development of investment fund industry.

an example of unacceptable risk; pushing the industry back by years. [How to respond to Bruno's various comments in the body of the paper]

The European Union's MiCA does not permit the issuance of algorithmic stablecoins, instead requiring that any significant stablecoin be fully reserved and subject to bank-like prudential rules. In practice, this means an algorithmic token with no issuer or no asset backing would not be allowed to call itself a "stablecoin" in the EU. Other regulators are issuing similar guidance. In the U.S, while no blanket ban exists yet, officials have repeatedly highlighted the failure of TerraUSD as justification for strict oversight of stablecoin arrangements that lack transparency or responsible management. However, this is not a closed case, as the broader crypto and CeFi industry still attempt to experiment with algorithmic stablecoins and have defended its future implementation(s).

It is likely that algorithmic stablecoins will either be forced into regulatory gray areas or develop into hybrid models with partial collateralization in order to comply with emerging standards, while this is not a proven model that is readily accepted by regulators, it could be a step in the right direction. From a policy standpoint, recent history has made the lesson clear: stable value promises require substantive assurances, and purely algorithmic stablecoins lack concrete provisions to guarantee asset stability and redeemability, whilst balancing economic incentives.

A.3 Exploring CBDC-Reserved Stablecoins

Another stablecoin category that surfaced during exploration of CBDC from a myriad of reserve and federal banks was CBDC-reserved stablecoins, stablecoins that are fully backed by central bank digital currencies. These assets have been positioned to alleviate the technical and regulatory constraints imposed by CBDCs, in a future where central banks issue retail or wholesale CBDCs, private intermediaries could effectively hold those risk-free digital reserves and issue their own stablecoin tokens against them. This model would blend the best of traditional banking systems and innovative programmability of smart contracts with a regulatory and technical sound front-end use case, abstracting the safety of public money as backing.

Several pilot projects have explored this design. Notably, the BIS Innovation Hub's Project Aurum (in collaboration with the Hong Kong Monetary Authority) prototyped a two-tier system featuring an "intermediated CBDC" alongside stablecoins backed 1:1 by CBDC held in the interbank system. (https://www.bis.org/about/bisih/topics/cbdc/rcbdc.htm) In such a framework, a commercial bank or licensed payment provider could issue a tokenized dollar (or other currency) to users, but every token in circulation is matched by an equivalent amount of central bank digital currency on deposit with the central bank.

CBDC-reserved stablecoins could address some of the key risks present in today's stablecoin arrangements. Credit risk and liquidity risk would be minimized, since the reserve asset is a direct claim on the central bank. It is feasible to see CBDC-reserved stablecoin virtually eliminate the danger of stablecoins due to reserve asset default or illiquidity, and could reassure users that their tokens are truly as good as central bank money. This would also simplify regulatory oversight of stablecoins, as authorities would primarily provide market assurance that the institutions issuing these types of stablecoins hold 100% reserves

コメントの追加 [30]: and VARA

コメントの追加 [31]: This section may be too long as a teaser section to the seusequent study. While it is very insightful to discuss this topic, we may need to touch on "who would be lienced to issue CBDC-Reserved Stablecoins? Most of developed markets, banks do not want to have other entities to have access to Wholesale CBDCs unless the entity obtain banking license. However, banks do not have any reason to issue CBDC-Reserved Stablecoins. Why banks market such stablecoins saying "this coin is much safer than the deposit account with my bank"?

Having said that, we can observe in some emerging market the central bank issue effective 'stablecoin' and allow payment service providers use it as backing asset. This is due to the lack of existing wholesale clearing systems among banks and PSPs.

コメントの追加 [32R31]: Agreed 18.6.25: this section can be a 'box' in the main body of the paper; it doesnt contribute to the Conclusion but adds an interesting dimension.

コメントの追加 [33]: Suggest reviewing the sytax of this sentence. I got lost what is S and V when reading.

コメントの追加 [34]: Very interesting. I read this paper several times in the past. I understand that is is quite similar to their current system that licenses HSBC and SC as the currency note issing bank in addition to HKMA itself. I heard the issuing business itself is not money making since they can't earn income from the reserved asset while it is very costly to pring and manage the distribution of the notes.

コメントの追加 [35]: To waht do you compare with? For example, Circle explain USDC is primarily backed by short term US Treasuries. How much are credit risks different between US Treasury and Fed Reserve? I agree liquidity risks are different.

コメントの追加 [36R35]: This is worth acknowledging as a live discussion, although we will not offer an analysis ourselves.

コメントの追加 [37]: Defaut of US Treasury? Fed does not fail?

コメントの追加 [38R37]: Different considerations for USD and non-USD currencies due to the USD's role as world reserve currency.

at all times. This would alter how stablecoin issuers fit into the financial system, as this model starts to resemble quasi-payment service providers, of which are providing a new front-end for distributing central bank money. Which would raise more policy considerations.

1. One issue would be that if stablecoin issuers hold large volumes of CBDC, they could become critical economic actors and have degrees of effect on monetary policy efficiency. This is already seen as an issue, with the expansion of US Dollar denominated stablecoins making up the overwhelming majority of available stablecoins. It is feasible to suggest that the proliferation of CBDC-reserved stablecoins would have a monumental impact as the economic line is tied between the reserve bank and the stablecoin issuer.

2. Another issue would be cross-border financial stability, as the asset would be hyper-mobile, which has the potential to complicate capital flow management and afford exacerbated responses during financial volatility.

Despite the many benefits and relative risks, it goes without much surprise that CBDC-reserved stablecoins blur the line between public and private money. In an instance they could be a way to harness private (especially web3) sector creativity to extend the reach of central bank money into new use cases, notable programmable contracts and lean financial institutions, something of which is readily developed and matured within DeFi, however, regulators would need to delineate responsibilities and ensure consumer protections.

It is equally understandable to expect central banks to impose strict requirements on any institution leveraging CBDC for a stablecoin, including operational and risk mitigation standards, redemption terms, and perhaps caps on volumes to manage systemic impact during volatile seasons. Legal frameworks might need updating to clarify that holders of a CBDC-backed stablecoin have a direct claim on the underlying central bank money (or equivalently, that the issuer's custody of the CBDC is bankruptcy-remote for the users' benefit) or if there is no claim, but rather direct redemption with the stablecoin issuer. While this stablecoin type is theoretical and viable, it will require clarity on the stance per jurisdiction on CBDC development, the desire for this asset to be available and regulatory frameworks on issuance and redemption to be solid before these assets become available, until then it will be valuable to continue following the various experiments in flight.

A.4 Identity & Stablecoins

- Bring IKP thought leadership to provide input
- USDC has functions to stop transactions with sanctioned entities - however, these are publicly available data
- Inter-bank use

A.5 Emerging Extraterritorial effects

It is clear that stablecoins are creating waves, as global traditional financial institutions experiment with new means of leveraging stablecoins innovation within global capital flows. Equally, authorities are placing a strong emphasis on protecting users and preserving

コメントの追加 [39]: Isn't it a mix of different phenomenon on monopolising the market. I read the former discusses oligopoly in CBDC market in one country and the latter in multicurrency global market.

コメントの追加 [40]: Agreed at WG meeting on 18/6: unless we are expecting specific input from IKP for this paper, this section should be removed, as it is out of scope.

コメントの追加 [41]: @mitchell@soulbis.com

Hey mate, can you provide some thought here from IKP?
Assigned to mitchell@soulbis.com

コメントの追加 [42R41]: :eyes:

financial sovereignty as these assets grow in volume and reach new borders. Stablecoins have for the most part been promoted for the opportunity to bring financial services to jurisdictions lacking financial infrastructure and offering alternative assets from local currency instability, for example, citizens in inflation-hit economies like Argentina and Turkey turned to USD-pegged stablecoins as a more reliable store of value (https://www.chainalysis.com/blog/2024-western-europe-crypto-adoption). While this can benefit consumers by preserving their purchasing power, it raises concerns for regulators about currency substitution. If households and businesses in an emerging market increasingly use a foreign stablecoin in lieu of the national currency, the central bank may find it harder to conduct monetary policy or act as lender of last resort.

Policymakers in jurisdictions facing this risk are exploring measures to maintain control, ranging from promoting viable central bank digital currencies or domestic stablecoin alternatives, to imposing limits or stricter oversight on the use of foreign stablecoins. Governments may also invoke capital flow management tools if they see large stablecoin flows as undermining foreign exchange regulations or enabling evasion of capital controls.

A.6 Protecting holders of coins

It should be noted that unlike bank deposits, stablecoin holdings generally do not benefit from insurance or guaranteed redemption at par by a central institution, dampening consumer protections against bank runs and similar financial crises, which could be further exacerbated if controls are not maintained within borders. This makes it vital for policy makers to create robust frameworks and standards to ensure holders have clear, enforceable rights and information, whilst affording access to new innovative means of global capital opportunities. IOSCO and the FSB have already urged that stablecoin arrangements be subject to the same rigor as traditional payment systems when it comes to reliability and consumer safeguards, given their potential role in everyday payments and cross-border transactions.

A.7 Arrangements recognizing the standards

Achieving interoperability and global standards is essential to realise the benefits of stablecoins. By design, stablecoins are borderless, a digital currency that can travel across the world as easily as an email; even faster in some cases. But if every country implements wholly different regulatory requirements (or technical standards) for stablecoins, this could create significant friction in what is otherwise would be a technically sound, auditable system. Asset holders might face situations where their origin or their current location stablecoin are legally recognised but however, if in another jurisdiction by either origin or current location may find that these stablecoins are unusable, or potentially illegal. Potentially in one jurisdiction transferring a stablecoin between two blockchains requires a third-party bridge, but this may be illegal or unsafe unless it is approved by the relevant authority.

It is possible for mutual recognition arrangements or passporting frameworks could be created to allow a stablecoin approved in one jurisdiction (under robust standards) to be more easily accepted elsewhere. This is analogous to how traditional financial institutions operate under home/host supervisor agreements. A coordinated global framework, even if

コメントの追加 [43]: Sustitute 'regulators' with 'goverments and central banks'

コメントの追加 [44]: how? any example? on custodial wallets?

コメントの追加 [45]: again, how?

コメントの追加 [46]: while it is an issue not to be covered by insurance (and LLR), it could be not suitable to compare with bank runs that are based on fractional reserve.

high-level, would help ensure that stablecoin users enjoy consistent protections regardless of where they are or which token they hold. In the interim, some countries are explicitly accounting for foreign stablecoins in their laws. Japan's law, for instance, will allow foreign-issued stablecoins to be traded in Japan only if they meet equivalently high standards of safety and transparency, thereby aligning with domestic consumer protection goals.(www.fsa.go.jp/en/news/2022/20220914-2/02.pdf)

5. Key Contributors

Chloe White (Lead Author), Independent Advisor, linktr.ee/chloe.white

6. References

https://link.springer.com/chapter/10.1007/978-3-031-48806-1_10